Assignment 3

Overview

Total: 100 points. Covers modules 06 (SVM and model evaluation), 07.1 and 07.2 (ensembles) and 08 (clustering). Individual work. Submit one Jupyter notebook and one PDF memo.

In this assignment you are the data scientist for a hotel group that runs a city hotel and a resort hotel in Portugal. Roughly 37% of their bookings get cancelled, which means empty rooms and lost revenue. The revenue team has asked you for two things: a model that predicts which bookings will cancel, so they know which ones to overbook against, and a segmentation of their guests, so marketing can design offers for different kinds of customers.

This is not a coding exercise. In every part you will be asked to make a decision, explain why you made it, and say what could go wrong. Plan on writing about as much text as code. If a question asks you to explain something and you only show output, you will get partial credit at best.

Data. `hotel_bookings.csv`, 119,390 bookings from July 2015 to August 2017 with 32 columns. Published by Antonio, de Almeida and Nunes (2019), Hotel booking demand datasets, Data in Brief. The target for Parts 1, 2 and 4 is `is_canceled`. Download link: https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-02-11/hotels.csv

Business numbers to use throughout. If the hotel overbooks a room and the guest shows up, walking the guest to another hotel costs 150 euros. If a booking cancels and the hotel did not overbook against it, the empty room costs on average 120 euros of lost revenue.

What to hand in

Item	Format	Notes
Notebook	`.ipynb` that runs from top to bottom after Restart and Run All	Put your reasoning in Markdown cells next to the code
Memo (Part 4)	PDF, two pages at most	Written for the revenue director, who does not know what AUC means
Seed	`random_state=42` everywhere	So we can reproduce your numbers

Keep the total run time under 15 minutes on a laptop. Deciding what not to compute is part of the job.

Part 1. Before you model (25 points)

The goal here is to set up an evaluation the business can trust before you train anything fancy.

1.1 Check the data and find the leak (10 points). Look at the dataset carefully and list, in a short table, every problem you find that could hurt a model. You should at least notice the missing values that carry information (what does it mean when `agent` or `company` is empty?), the strange values in `adr`, and the bookings with no guests at all. For each one, say what you did about it and why. "Dropped the rows" with no reason gets no credit.

Then find the leak. Two columns in this file would let any model score almost perfectly, and both would be useless in practice. Identify them, show the evidence in one table, explain in plain business language why a model built on them cannot be deployed, and remove them before going further.

1.2 Cost matrix and the two trivial policies (7 points). Using the business numbers above, write out the 2 by 2 cost matrix for predicting "will cancel" versus "will not cancel". Say clearly which cell gets which euro amount. Then compute, on your training set, the total cost of two policies that need no model at all: overbook against nobody and overbook against everybody. Any model you build later must beat these numbers, so keep them.

1.3 Logistic regression and SVM (8 points). Split off a holdout test set and do not touch it again until Part 4. Train a logistic regression and an RBF kernel SVM, each inside a pipeline with the preprocessing it needs. Evaluate both with stratified 5-fold cross validation on the training set and report AUC, recall, precision and total cost at the threshold that minimizes cost (not at 0.5). Explain in a couple of sentences why the best threshold is not 0.5.

The SVM will not finish on 90,000 rows in any reasonable time. Decide how to deal with that, for example by training on a stratified subsample, and show with a quick experiment how much AUC you give up. This is a real constraint you will meet at work, not a flaw in the assignment.

Part 2. Ensembles (30 points)

In Modules 07.1 and 07.2. The lectures made one central claim: bagging reduces variance and boosting reduces bias. Here you check whether you can tell which problem a model has before you pick the fix.

2.1 Diagnose a single tree (8 points). Train decision trees with `max_depth` set to 2, 4, 8, 16 and unlimited. Plot training AUC and 5-fold CV AUC against depth. For the depth 2 tree and the unlimited tree, say in one sentence each whether the problem is bias or variance. Then, before you run anything else, write down your prediction: which of bagging or boosting will help each of these two trees more?

2.2 Test the prediction (10 points). Train a bagging ensemble and a gradient boosting ensemble (sklearn or XGBoost) using each of the two trees from 2.1 as the base learner, so four models in total. Also train a random forest with 200 trees. Report CV AUC and cost at the best threshold for all five. Compare the results with the prediction you wrote in 2.1 and explain anything that surprised you. A wrong prediction with an honest and correct explanation gets the same marks as a right one.

2.3 What drives cancellations (12 points). Take your best boosted model and list the ten most important features. Then do the part a data scientist is actually paid for. Turn the top five into plain English statements that the revenue director could act on. One of them will look odd: in this data, bookings with a non refundable deposit cancel far more often than the others. Before you report that, dig into it. Which market segments and agents are those bookings coming from? Do you believe the hotel could reduce cancellations by changing its deposit policy, or is something else going on? Finally, compare your top ten with the top ten from the random forest and explain why two good models can rank the same features differently.

Part 3. Who are our guests (25 points)

Marketing wants three to six guest segments they can name and target. There is no label and no accuracy score to hide behind, so the job is to produce segments that are stable and make sense, and to be honest about where they do not.

Use only bookings that were not cancelled and take a random sample of 10,000 of them. Start from these features: lead_time, stays_in_weekend_nights, stays_in_week_nights, adults, children, adr, total_of_special_requests, booking_changes, previous_bookings_not_canceled. You can add or remove features if you explain why.

3.1 Scale the data and choose k (10 points). Run k means with k = 4 once on the raw features and once on standardised features. Show which feature takes over the raw clustering and explain, using the distance formula from the lecture, why that happens. Look at the distributions of adr and lead_time and decide whether standardising is enough or whether a log transform first would be better. Then produce the elbow plot and the average silhouette plot for k from 2 to 10. State what k each one suggests. They may not agree. Explain what each method actually measures and pick a k that marketing can use, with a reason better than "the plot said so". Run k means at your chosen k with at least 10 different initialisations and report how much the inertia moved between runs.

3.2 Name the segments (8 points). Build a one row profile for each cluster: size, mean or median of each feature, share of resort hotel bookings, and the most common market_segment. Give each cluster a name a marketer would understand (something like "last minute weekend couples", not "cluster 2") and one concrete offer you would design for it. Then point out the cluster you trust least and say why. Finally, assign the cancelled bookings to their nearest centroid and cross tabulate cluster against is_canceled. Does any segment cancel a lot more than the others? What does that tell the revenue team that Part 2 did not?

3.3 Are the groups real (7 points). Run DBSCAN on the same standardised features. Pick eps from the k distance plot, using MinPts = 2 times the number of features as the lecture suggests, and report how many clusters and how many noise points you get. Then compute the adjusted Rand index between your k means labels and the DBSCAN labels, treating noise as its own label, and interpret it. Finish with a short answer to the real question: do these guests fall into natural groups, or did k means just cut a continuous cloud into k pieces? Use the silhouette values, the DBSCAN result and one 2D plot (PCA or two features of your choice) as evidence. Either answer can get full marks if it is supported. End with one sentence on what this means for how marketing should treat the segment names.

Part 4. The final test and the memo (20 points)

This is the part the hotel actually pays for. Everything before it was preparation.

4.1 Test the model, twice (8 points). Choose your final cancellation model and threshold using only the cross-validation results from Parts 1 and 2. Refit on the full training set, score the hold out test set once, and report AUC, recall, precision and total cost next to the two trivial policies from 1.2. If the test cost is noticeably worse than the CV cost, say so and give the most likely reason.

Your test set was a random split, but the hotel will run this model on future bookings. So repeat the evaluation with a time-based split: train on arrivals in 2015 and 2016, test on 2017. Compare the two test costs. Which number should the revenue director believe, and why? Find one thing in the data that could make 2017 behave differently from 2016. Look for it, do not guess.

4.2 Memo to the revenue director (12 points). PDF, two pages maximum. It needs:

1. The recommendation in the first two sentences: which bookings to overbook against, and how much this saves per month compared with overbooking against nobody. Say it in euros and in rooms, never in AUC.

2. One chart that a non technical reader understands in ten seconds. Think about which one. A ROC curve is almost certainly the wrong choice.
3. The guest segments from Part 3 as a small table with your names and one offer each, plus one honest sentence about how confident you are that the groups are real.
4. Three risks of acting on the model, each with something you would do about it. At least one should be about the data (what happens if the hotel changes its deposit policy?) and one about the people affected (the guest who gets walked).
5. One specific piece of data the hotel is not collecting that would most improve the model, and why.

The memo is marked on judgement and clarity, not length. If the director could forward it to the CEO without editing, that is full marks. Technical jargon will cost points, not earn them.

You may use AI tools for code, the way you would at work. You cannot use them to write your explanations, your predictions or the memo. Those are the parts being graded, and they are the parts an employer cannot outsource either. If you used an AI tool, say so in one Markdown cell at the top of the notebook. Be ready to explain any line of your notebook in a short conversation if asked.